



A cyber risk prediction model using common vulnerabilities and exposures

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ABSTRACT

The cyber risk from malicious external attackers is a significant socio-economic problem. Cyber risk prediction is particularly difficult, given the constantly changing attack vectors. This study presents a model that automatically predicts cyber risks. The model is only based on common vulnerabilities and exposures (CVE) data and supervised prediction algorithms. This approach eliminates expert opinion bias in cyber risk prediction. Our supervised data-driven model, *CyRiPred*, CVE data into cyber risk groups by mapping the textual description field of the database into relevant Wikipedia article titles. Then *CyRiPred* aggregates the occurrence and severity of extracted topics for the desired time unit and produces a time series fed to supervised regressors for prediction. The risks are calculated using predicted occurrence and impact. Finally, the cyber risks are ranked by their score, and the top ten risks are presented. The proposed model is evaluated, and the results are discussed.

1. Introduction

Cyber risk targets software or hardware systems, whether stand-alone, distributed, or network architecture. Identifying and assessing cyber risks is vital for information system users, IT-based organisations, cybersecurity experts, and cyber risk insurers. Assessment and Identification of cyber risk require developing special techniques and procedures (Refsdal et al., 2015; Sheehan et al., 2021). To underwrite the risks, the insurer needs to understand the risks and the impact of a cyber-attack. A risk impact score depends on the impact and likelihood of risk occurrence (Alali et al., 2018). There is limited recorded information for academic research, and most predictions depend on experts' knowledge and subjective understanding of current and future risk scenarios.

When predicting future cyber risk, the insurance industry and cybersecurity companies produce cyber risk prediction and trends reports. However, the methodology used to create these reports is based on expert opinion, and annual reports are not sufficiently frequent. IBM, Norton, and many other companies provide these reports (Rawlings, 2015). While companies and research groups work on this area with different methodologies, there is still a gap between a reliable, comprehensive, real-time, data-driven method and their research.

There are many different approaches to looking at the problem in the cyber risk prediction literature. These approaches can be distinguished into two general categories: The first category includes those researches that try to predict cyber incidents and risks inside a specific individual system or network (Khodabakhsh et al., 2020; Sentuna et al.,

2021). These researches are used to improve the efficiency of Intrusion Detection Systems (Buczak & Guven, 2016) by predicting a specific type of failure before it happens (Salfner et al., 2010). The second category is those the researches that are environment-independent and investigate the concept of cyber risk regardless of focusing on any particular application, system, or network ecosystem (Subroto & Apriyana, 2019). Our model is similar in the second category. However, to our knowledge, there is no peer-reviewed comprehensive study that is both environment-independent and application-independent and is entirely data-driven on the Common Vulnerabilities and Exposures dataset.

The ability to forecast cyber attacks will significantly limit the socio-economic impacts of such events. In our model, which we call *CyRiPred* (Cyber Risk Prediction Model), whenever a new vulnerability comes in the National Vulnerabilities Database (NVD), our system immediately associates that risk to a related cybersecurity topic using our novel *WikiTopic* sub-module, which uses a keyword extraction algorithm and a consecutive Wikipedia search. We further assign a severity score to each topic using aggregated impact scores in the CVE database. The CVE database gathers publicly known cybersecurity vulnerabilities and exposures (Cremer et al., 2022). Finally, we apply the aggregated cyber risk topic occurrences and impact scores to historical data to predict future emerging risk topics.

In general, research that identifies cyber risks lacks a method to generalise the CVEs into more meaningful groups. For example, many vulnerabilities are related to Microsoft Windows with different CVE codes. There were more than 140 K CVEs up to the year 2020. This

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highly granular data inhibits an analysis leading to a more general understanding of cyber vulnerabilities' underlying structure. A famous effort to categorise CVEs into more general groups is Common Weakness Enumeration (Christey et al., 2013). However, besides helping define a categorisation platform, Common Weakness Enumeration (CWE) has been criticised as being inaccurate, incomplete, inconsistent, or ambiguous (Black et al., 2015). There were more than 600 CWEs up to the year 2020.

To our knowledge, none of the models built to predict cyber risk are comprehensive. Each model pays attention to one dimension of cyber risk or a specific type. Chen et al. (2018) tried to forecast if a new CVE would be rejected by using its records as input features and Random Forest (RF), SVM, and Naïve Bayes (NB) models as binary forecasters. Bilge et al. (2017) used binary files data inside a network to discover new cyber risks. Schultz et al. (2020) built a system to forecast damages on network assets using probabilistic models. Application-based cyber risk prediction models for IoT, Connected and Autonomous Vehicles (CAV), etc., have also been mentioned.

In risk prediction and system protection, leveraging a 'first principles' methodology ensures that foundational truths and basic concepts are at the heart of the analysis. One can derive more reliable and insightful conclusions by dissecting complex systems into their elementary components and understanding their inherent properties and interrelations. A notable application of this approach is seen in the work of Chai et al. (2011), where they employ the first principles combined with social network theory to prioritise protection in the interconnected oil and gas industries within a networked critical infrastructure system. Their study breaks down the intricate web of industry interactions into elemental relationships, emphasising the fundamental dynamics of the network. Drawing inspiration from such a methodology, our model, *CyRiPred*, seeks to predict cyber risks using a data-driven approach rooted in the foundational data of Common Vulnerabilities and Exposures (CVE), circumventing the biases often inherent in expert-driven analyses.

Our model identifies a cyber risk topic in real-time, meaning that the model extracts the related cybersecurity topic whenever a new vulnerability comes into the National Vulnerabilities Database (NVD). This feature in our model differentiates it from the CWE, whose shortcomings are overcome by our novel *WikiTopic* topic extraction algorithm that uses CVE categorisation in cyber risk groups. Using an open-source knowledge repository like Wikipedia also has the advantage of achieving a higher resolution for cyber risks compared to CWE, which has just over 600 enumerations. Wikipedia had been used as a topic repository in non-explicit topic extractors (Yun et al., 2011) and in topic mapping to video data (Roy et al., 2011). Wikipedia is the most extensive knowledge repository and the most comprehensive ontology resource at hand (Ensslin, 2011; Miz et al., 2017; Pilkauskas, 2010; Sinanc & Yavanoglu, 2013). The authors know that Wikipedia is not an academic peer-reviewed encyclopedia of cybersecurity topics. However, it is the best open-source knowledge repository at hand that is used in many intelligent NLP-based tools in different expert domains (del Valle et al., 2018; Lämmel et al., 2013; Noraset et al., 2021; Schultz et al., 2000). After retrieving the topics, the likelihood of topic communities' occurrences and their severity from CVSS are extracted for each time unit. Then, the extracted time series of cyber risk topics occurrences and impacts are used to calculate the time series's risk score. Finally, the computed series are fed to ML prediction models. The model can update its prediction with every instant update in the NVD.

Our study provides a model for categorising the types of cyber risks as well as predicting the future of these risks based on an automated data-driven method. The model only requires CVE data, and then it identifies and predicts cyber risks automatically with novel Natural Language Processing (NLP) and Machine Learning (ML) algorithms. Cyber-attacks constantly change, and there is a societal and market demand for forward-looking early-warning systems (Kopp et al., 2017).

In the next section, a general overview of the model is presented. The *WikiTopic* algorithm is explained in detail, describing the risk calculation and forecast models. Section 3 discusses the CVE dataset and offers some of its statistics. In Section 4, the results of the *CyRiPred* are presented. Section 5 evaluates the model and all its modules and *WikiTopic* algorithm. In Section 6, we discuss the results produced by our novel *CyRiPred* model before a final concluding section.

2. Methodology

2.1. General overview of *CyRiPred*

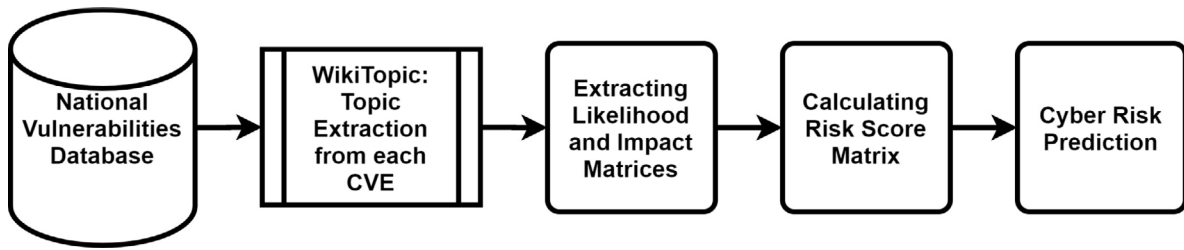
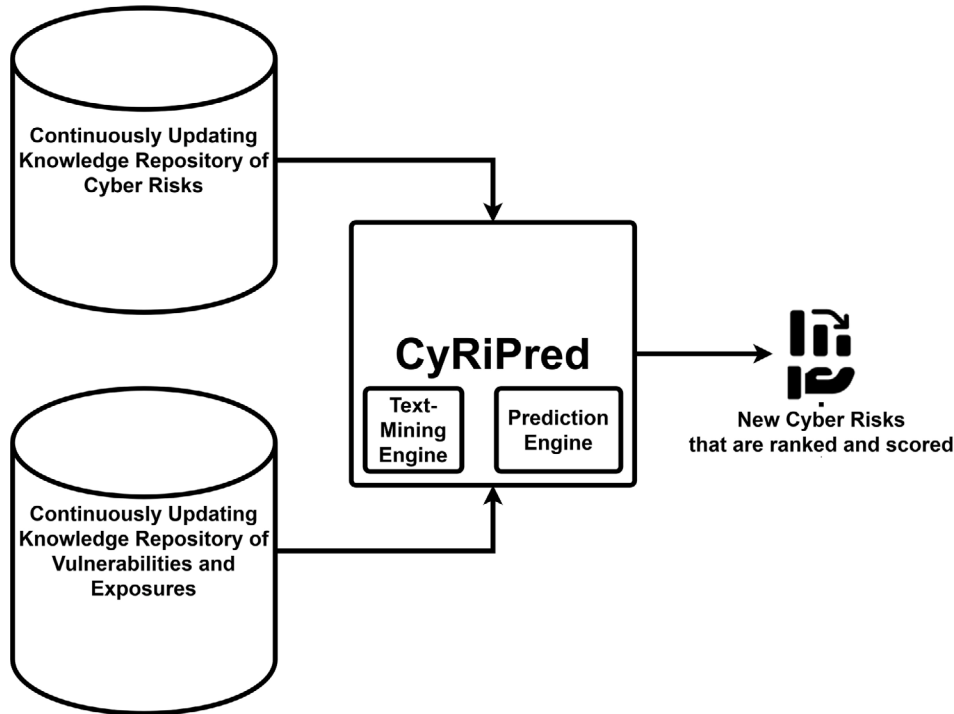
CyRiPred stands for cyber risk recognition and prediction model. The presented model is data-driven, taking historical data on cybersecurity vulnerabilities and exposures as input and predicting future cyber risks. CVEs in the NVD dataset are used as input. In our proposed methodology, for each CVE in NVD, we focus on three fields: a text field that describes the vulnerability or exposure, a numerical field called base score and records the severity of the vulnerability or exposure, and a date field that shows the first time the vulnerability or exposure was discovered. A comprehensive explanation of the dataset used in the proposed method is presented in Section 3.

Fig. 1 presents the general steps of our proposed methodology. In this section, each step – each box in Fig. 1 – is explained in detail. Initially, each description field in each CVE will be entered as input to a topic extraction algorithm to be labelled with a cybersecurity topic. In this way, significant groups of cybersecurity problems are recognised. The next step is to calculate the impact and likelihood of the occurrence of cybersecurity topics for each month and year (or any other time unit). We need to calculate the likelihood and impact of each cybersecurity topic to calculate the risk score. Risk is calculated as in Eq. (1).

$$\text{Risk} = \text{Likelihood} \times \text{Impact} \quad (1)$$

The likelihood of the occurrence of a cybersecurity topic is the total number of specific cybersecurity topics per time unit. The CVSS base score can be considered as an indicator of the impact of a CVE (Fruh-wirth & Mannisto, 2009; Houmb et al., 2010). We calculate the average of base Scores for each cybersecurity topic as the impact in risk calculation. Finally, we will have each cybersecurity topic's time series of occurrence, impact, and risk score. By entering these time series as the input of a prediction model, we can produce a prediction for the next time unit (e.g., month or year).

Our model identifies known and new risks from an updating knowledge repository (in our case, Wikipedia) by assigning risk labels (entry topics) to new vulnerabilities and exposures from an updating dataset of vulnerabilities (in our case, CVE dataset) with the help of using a novel text-mining algorithm. Our model can attach to two updating knowledge repositories (One with risk labels and one with vulnerabilities and exposures) and identify and rank known and future cyber risks. The model predicts future risks based on the discovered patterns of occurrences and impacts of identified risks from the knowledge repository (Wikipedia in our case). The overview of *CyRiPred* implementation is also presented graphically in Fig. 2. Here we should mention an important issue: In prediction research, there are two types of future risks: One that has been seen in the past or now or we have a trace of it in our data and those that have not been seen before. The nature of Machine Learning models is data-driven. They assume that the future can be predicted based on past historical data. The model cannot predict if a pattern has been seen in the historical data. Machine Learning methods are designed to forecast the future based on past data (they are data-driven). They cannot foresight a risk that has not been seen before, and there is no trace of it in the historical data. This can be seen as a limitation of data-driven machine-learning models. However, our novel model uses updating knowledge repositories for risk labelling and vulnerabilities and exposures. This helps our model capture new risks with an entry/label in our knowledge repository (Wikipedia).

Fig. 1. General steps of the *CyRiPred* methodology.Fig. 2. An overview of how *CyRiPred* is implemented.

2.2. WikiTopic: A novel explicit topic extraction algorithm

In the first step, a descriptive textual field of each CVE is entered into a topic extraction algorithm that explicitly assigns each CVE to a cybersecurity topic. All CVEs with the same cybersecurity topic are put into the same category. These categories can be considered clusters in unsupervised learning methods.

Most existing topic modelling algorithms provide multiple topics as output, where each topic contains various words, and each word in the topic has a probability score showing its importance in the extracted topic (Alghamdi & Alfalqi, 2015; Daud et al., 2010; Jelodar et al., 2019; Padmaja et al., 2018). These algorithms have statistical assumptions, for example, about the distribution of words and topics. In these algorithms, the algorithm user must set the number of issues. In our study, we do not assume prior knowledge about the number of cybersecurity topics that can be extracted from the NDV database. The most common topic modelling algorithms are Latent Semantic Analysis (LSA), Probabilistic Latent Semantic Analysis (PLSA), and Latent Dirichlet allocation (LDA) (Blei & Lafferty, 2006; Blei et al., 2003; Dumais, 2004). These models require extensive development and modification to work with short texts (Qiang et al., 2020). Some other algorithms developed for short texts have the same drawbacks of too many parameters to adjust and default statistical assumptions about word frequency distribution in the corpus. These issues with the conventional topic modelling algorithms and the requirement for an

algorithm that explicitly outputs one topic per CVE led us to develop our explicit topic extraction method called *WikiTopic*.

In the *WikiTopic* algorithm, Wikipedia is used as a comprehensive open-source human-knowledge repository of topics. As a data source, Wikipedia offers advantages like a balanced perspective, high reliability due to constant editing, and comprehensive coverage of topics. Meanwhile, social media can sometimes introduce biases due to user sentiment or localised events. . Wikipedia topics have been used in research to extend LDA, but the problems mentioned before with LDA, such as assigning the number of topics and distribution assumption, remain (Hansen et al., 2013). Our approach is simple; we want to assign the topic of the most similar Wikipedia article to the input text. The key here is the definition of similarity. To find the most similar Wikipedia article to a text, we used the search API provided by Wikipedia and keywords extracted from the text by a keyword extraction algorithm. For keyword extraction, we use the Yet Another Keyword Extraction (YAKE) algorithm that performs better than other keyword extraction algorithms (Campos et al., 2018a, 2018b, 2020). YAKE is a statistical unsupervised keyword extraction algorithm that uses five different features for each term in text and calculates a keyword candidate score. Then, it outputs keywords by the computed scores. For further details of the YAKE, the interested reader can read (Campos et al., 2020).

Fig. 3 shows the flowchart of the proposed *WikiTopic* algorithm. At first, the input text should be prepared by lemmatising and removing the stop-words. Lemmatising is the process of extracting each word's

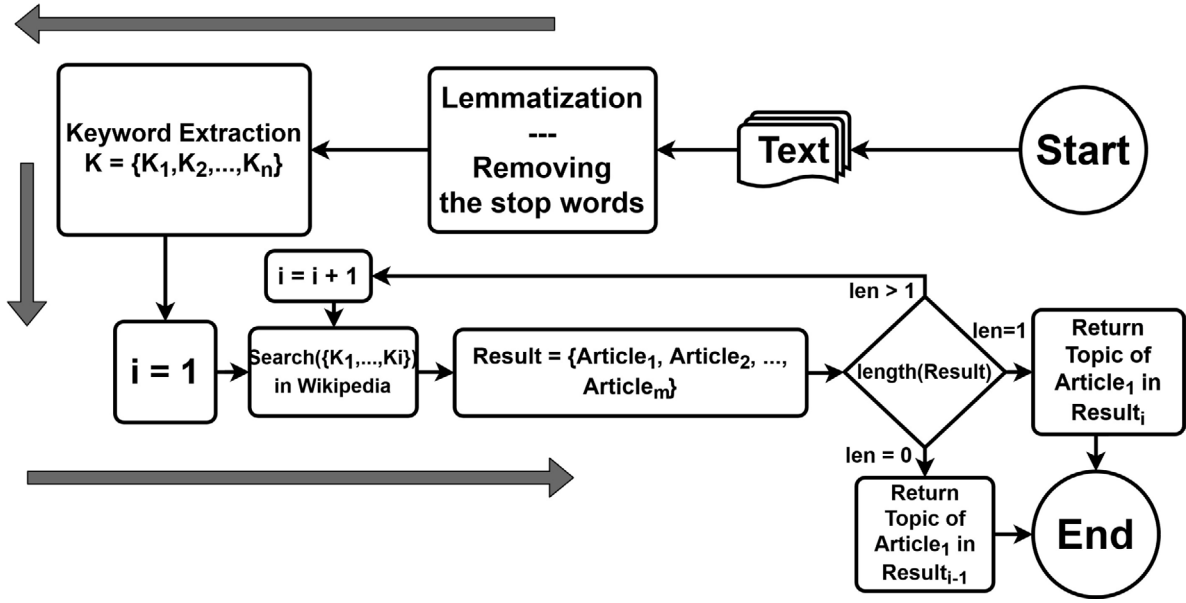


Fig. 3. Flowchart of WikiTopic explicit topic extraction algorithm.

root in the text and replacing it with the actual word. Stop words are the most common words used in a language. Examples of stop-words include “a, an, the, of” and other common words. The lemma, which is also called the non-inflected or dictionary form of the word, is also a better input for many NLP applications and can increase the performance of keyword extraction algorithm and Wikipedia search API and reduce confusion that may happen with different inflected forms of a word (Müller et al., 2015).

The data preparation (Lemmatization and stop-word elimination) produces a list of keywords for the input text. Assume $K = \{K_1, K_2, \dots, K_n\}$ are n different keywords extracted from the text where their importance as a keyword sorts them in decreasing order. The keyword importance score comes from the YAKE algorithm, which takes Term Frequency – Inverse Document Frequency (TF-IDF) and other variables into account (Campos et al., 2020). This means that K_1 is the most important keyword in the text, and K_n is the least important one. The WikiTopic algorithm uses iteration to search Wikipedia with an incremental list of $K' = \{K_1, K_2, \dots, K_i\}$. In the first iteration of WikiTopic, the Wikipedia articles are searched with the K_1 keyword. Then, the Wikipedia API returns a list of articles as the search output with this keyword. In the second iteration of the WikiTopic, the Wikipedia is searched with K_1 and K_2 . The iteration continues until the search result list becomes empty or only one article. Adding keywords to the search input narrows the search until we only have one article or no articles. If the search result's length is one, then the article's topic returned from the search will be the topic of the input text. Suppose the search result becomes empty in an iteration. In that case, the first search result of the previous iteration will be considered the closest article to the input text, and its topic will be returned as the output of WikiTopic. The WikiTopic is available to readers as an open-source Python package in PyPi and GitHub under GNU General Public License version v3.0.¹

After running the WikiTopic on all CVEs, we will have a new field (feature) in our dataset, the cybersecurity topic for each CVE. Eq. (2) shows the list of CVEs and the list of topics as W_1 to W_p . Assume that we will have q different cybersecurity topics in the Topics list. Each of these cybersecurity topics will correspond to a column in Occurrence,

baseScore, and Risk matrices in the next section.

$$CVES = \{CVE_1, CVE_2, \dots, CVE_p\}$$

$$Cybersecurity\ Topics = WikiTopic(CVES) = \{W_1, W_2, \dots, W_p\} \quad (2)$$

$$Distinctive\ Cybersecurity\ Topics = T_1, T_2, \dots, T_q$$

2.3. Likelihood, impact, and risk calculation

Given q specific cybersecurity topics as shown in Eq. (2), we make two matrices with $t \times q$ dimension where t is the total number of data in each column (each time series of topic occurrence and baseScore for each cybersecurity topic). The time unit of each time series (each column in matrices of Eq. (3)) can be day, month, or year, depending on the aggregation method used on the date feature of the primary CVE dataset. There will be a comprehensive explanation of the CVE dataset in Section 3, where we will talk about the dataset used as the input of our CyRiPred model.

$$\begin{aligned} \text{Topic Occurrence Matrix} &= \begin{bmatrix} T_{1,1} & \dots & T_{1,q} \\ \vdots & \ddots & \vdots \\ T_{t,1} & \dots & T_{t,q} \end{bmatrix} \\ \text{baseScore Matrix} &= \begin{bmatrix} B_{1,1} & \dots & B_{1,q} \\ \vdots & \ddots & \vdots \\ B_{t,1} & \dots & B_{t,q} \end{bmatrix} \end{aligned} \quad (3)$$

From Eq. (3) and Topic Occurrence Matrix, $T_{i,j}$ is the number of occurrences for cybersecurity topic j in time i . In the baseScore Matrix, $B_{i,j}$ is the average base Score for cybersecurity topic j in time i . The risk equation will be written as Eq. (4), where the inner product of two matrices of Occurrence and baseScore will result in the risk score matrix.

$$\text{RiskScore} = \text{Topic Occurrence} \bullet \text{baseScore} \quad (\text{innerproduct}) \quad (4)$$

The risk score matrix is presented in Eq. (5) where $R_{i,j}$ is the risk score for cyber risk related to cybersecurity topic j in time i .

$$\text{Risk Score Matrix} = \begin{bmatrix} T_{1,1}B_{1,1} & \dots & T_{1,q}B_{1,q} \\ \vdots & \ddots & \vdots \\ T_{t,1}B_{t,1} & \dots & T_{t,q}B_{t,q} \end{bmatrix} = \begin{bmatrix} R_{1,1} & \dots & R_{1,q} \\ \vdots & \ddots & \vdots \\ R_{t,1} & \dots & R_{t,q} \end{bmatrix} \quad (5)$$

In Fig. 4, a glimpse of what happens in actual data and results is presented to view better the significance of Eq. (5). The column

¹ WikiTopic Python Source Code and Package Information: <https://github.com/ConKruG/WikiTopic>

Occurrence (Frequency)					×	baseScore (Impact)					=	Risk Score				
Date	Transport Layer Security	Cross-site scripting	...	SQL injection		Date	Transport Layer Security	Cross-site scripting	...	SQL injection		Date	Transport Layer Security	Cross-site scripting	...	SQL injection
2000	12	2	...	0		2000	5.94	8.75	...	No Data		2000	71.30	17.50	...	0.00
2001	18	12	...	7		2001	6.31	6.53	...	6.67		2001	113.60	78.40	...	46.70
⋮	⋮	⋮	...	⋮		⋮	⋮	⋮	...	⋮		⋮	⋮	⋮	...	⋮
2010	273	166	...	212	×	2010	6.59	4.68	...	7.02	=	2010	1798.40	777.50	...	1487.30
⋮	⋮	⋮	...	⋮		⋮	⋮	⋮	...	⋮		⋮	⋮	⋮	...	⋮
2019	226	155	...	207		2019	5.18	4.48	...	5.27		2019	1170.90	694.90	...	1090.00
2020	140	105	...	150		2020	5.23	4.55	...	5.27		2020	732.67	477.65	...	790.54
2021	Prediction	Prediction	...	Prediction		2021	Prediction	Prediction	...	Prediction		2021	Prediction	Prediction	...	Prediction

Fig. 4. Cyber risk score calculation after topic extraction and likelihood (Frequency or Occurrence) and baseScore (Impact) calculation. Transport Layer Security, Cross-site scripting, and SQL injection are three samples of topics extracted from the CVE description field with the *WikiTopic* Algorithm. The multiplication operator in the figure is an inner product (Multiplication element by element).

		Severity			
		1	2	3	
Risk Classification Matrix	=	Low	Low	Medium	1
		Low	Medium	High	2
		Medium	High	High	3

Fig. 5. Risk classification matrix.

headers are some of the topics extracted from CVEs description fields. These topics are common for many CVE descriptions and, therefore, make clusters of CVEs under the name of a cybersecurity topic. Let us consider each column as a time series for the likelihood (frequency), *baseScore* (impact), and the cyber risk score of each cybersecurity topic. We can use a prediction model to predict the future values for these time series.

Before discussing the prediction phase, it is instructive to consider the possibility of classifying cyber risks under the *CyRiPred* model. Dividing the intervals of continuous values for occurrence and *baseScore*, risk can be classified with a risk classification matrix such as Fig. 5. According to NIST (Information Technology Laboratory, 2020), in some applications, the *baseScore* is discretised by dividing it into three or five intervals. For example, in the three-interval division, the *baseScore* interval from 0 to 3.99 indicates low vulnerability or exposure severity, 3.99 to 7 indicates medium severity, and 7 to 10, high severity. The different discretisation of the continuous values of *baseScore* is presented in NIST-NVD online documents (ibid). The likelihood of a cybersecurity topic in our work is the number of occurrences of CVEs with that topic in a period. We can scale all the likelihood data into any interval and then discretise it the same as the *baseScore*. After discretising *baseScore* and Topic Occurrence, we can use a risk classification matrix to classify cybersecurity risks extracted under *WikiTopic*, CVE clustering, and *baseScore*, *TopicOccurrence* aggregation for the desired period (desired time unit).

2.4. Cyber risk prediction

Each column of *Occurrence*, *baseScore*, and *CyberRisk* matrices in Eqs. (3), and (5) and in Fig. 4 are a time series of a cybersecurity topics likelihood, impact, and risk. For example, the time series of cybersecurity risk *i* is presented in Eq. (6). The *PredictionModel* in Eq. (6) can be any regressor with a tolerable error in the application domain. Our model's cyber risk score is computed by predicting the risk time series for each cyber risk topic, as explained in Eq. (6). The cyber risks are first calculated using known occurrence and impact data, and then future risks are predicted. The minimum acceptable performance is a function of business conditions and the maximum prediction performance that business experts can provide without using an intelligent system. It may also be possible that each time series (each column in one of the three matrices for likelihood, impact, and risk corresponding to a different

cybersecurity topic) can be more accurately predicted with a different regressor.

$$\begin{aligned} S_i &= R_{1,i}, R_{2,i}, \dots, R_{t,i} \\ R_{t+1,i} &= \text{PredictionModel}(S_i) \end{aligned} \quad (6)$$

After predicting all the values for the next year or month for likelihood, impact, and cyber risk score for cybersecurity topics, we classify each cyber risk with the risk matrix classification method in Fig. 5 or sort the cybersecurity topics to find out top-ranking future cybersecurity issues. In Section 4, we find a random forest regressor had the best performance in terms of average accuracy for all the extracted cybersecurity topics. Random forest is an ensemble machine learning model that consists of a set of decision trees on random subsets of a dataset (Ho, 1998). Each decision tree is trained on a random subset of a delay-embedded time series in our study. It should be mentioned that the time series should be converted to a delay-embedded format before being fed to the random forest. In a delay-embedded format, each time series will convert to a table where each row will have *n* consecutive elements, and the last element will be used as the training label. Random forest randomly selects several rows and columns of the delay-embedded time series and creates random subsets as new datasets. Finally, a decision tree regressor is trained for each subset. Each decision tree regressor gives us a prediction. The average prediction of all decision trees on all random subsets of data is the final prediction of the random forest model. It is important to mention that there might be correlation between different cyber risks that may provide an opportunity to predict multi-variate time series to capture these interrelationships. However, due to lack of enough data to capture the interrelationship between all the extracted time series and simplicity of the methodology using uni-variate time series prediction, we stuck with the univariate approach. We acknowledging the potential for future work in multivariate approaches.

3. Data

NVD presents data in both Javascript Object Notation (JSON) and eXtensible Markup Language (XML) format.² There are many different fields in the database for each CVE. Among all the features, we used three fields: Description (to use in topic extraction algorithm and label a topic to each CVE), *baseScore* (to calculate the severity or impact of each cybersecurity topic group), and Date (to use in averaging and summation aggregation and sequencing the data in time series used in prediction models). These fields and two other fields are shown in Table 1. For clarity, we added two fields, a unique CVE code and a qualitative discretised field called Severity that labels each CVE into three classes of Low, Medium, and High according to their *baseScore*. The last column of Table 1 is the topic extracted from the Description field after running the *WikiTopic* algorithm on the dataset.

² [dataset] CVE database: [lastaccessed:1thJan.2021](#)

Table 1

A glimpse of the dataset used in *CyRiPred* model. The results of the *WikiTopic* algorithm are presented in the “Retracted Topic with WikiTopic” field. These results should be read as “Related to PHP” or “Related to DragonFly BSD”. A “Related to” is better to put at the beginning of all the topic results from the *WikiTopic* so that it makes sense.

Date	CVE code	Description	baseScore	Severity	Extracted topic with <i>WikiTopic</i>
10/1/1988	CVE-1999-0095	The debug command in Sendmail is enabled, allowing attackers to execute commands as root.	10	HIGH	DragonFly BSD
⋮	⋮	⋮	⋮	⋮	⋮
2/8/2012	CVE-2012-1023	Open redirect vulnerability in admin/index.php in 4images 1.7.10 allows remote attackers to redirect users to arbitrary websites and conduct phishing attacks via a URL in the redirect parameter.	5.8	MEDIUM	PHP
⋮	⋮	⋮	⋮	⋮	⋮
9/1/2020	CVE-2020-6137	SQL injection vulnerability exists in the password reset functionality of OS4Ed openSIS 7.3. The password_stf_email parameter in the password reset page /opensis/ResetUserInfo.php is vulnerable to SQL injection. An attacker can send an HTTP request to trigger this vulnerability.	7.5	HIGH	Password cracking
⋮	⋮	⋮	⋮	⋮	⋮

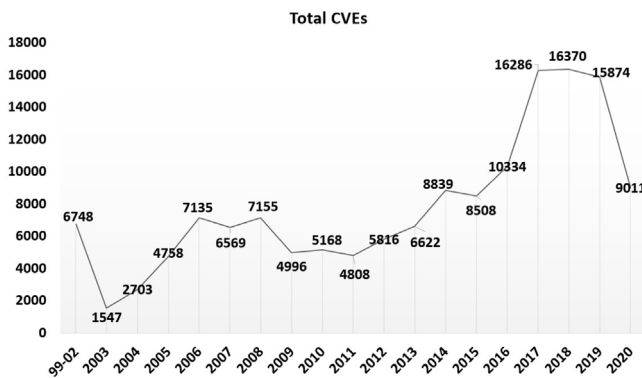


Fig. 6. Total number of CVEs for each year.

There are 149,247 CVEs in our dataset. After eliminating the rejected/disputed data labelled in the NVD database, 140,213 CVEs remain. The first CVE was from 1988 and was recorded in NVD in 1999. The last CVE in our dataset is for 4th September 2020. The NVD is updated daily with new CVEs.

Fig. 6 shows graphs of total CVEs per year. The downward movement for 2020 is due to our dataset's not having data for the months after September. An increasing pattern is seen in the recorded CVEs over the years.

Fig. 7 shows the total number of CVEs with three different severity classes in NVD. The graph shows more increase in the high severity class than two other classes in recent years.

4. Results

In this section, the likelihood (number of occurrences of a cybersecurity topic group), the impact severity (average baseScore for a topic group), and the cybersecurity risk scores calculated for the extracted topic groups are ranked for each year and all years. There were 20,181 different cybersecurity topics extracted from the dataset, with 7875 topics with more than one occurrence. The top ten rankings for likelihood, impact, and cyber risk and the predictions for next year

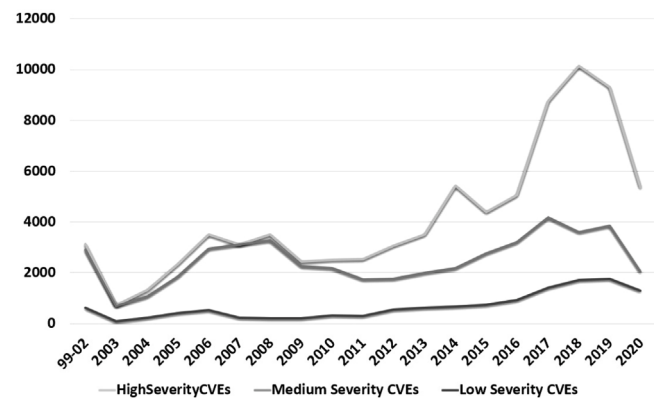


Fig. 7. Total number of severity classes recorded for CVEs per year.

(2021) are then presented as an important part of the results made by *CyRiPred*. For some of the results (the first three cybersecurity topic groups in each table), we will briefly explain the topic itself. However, for further studies and more information on the technical details of computer networks and security, the reader can refer to [Peterson and Davie \(2007\)](#) and [Gollmann \(2010\)](#).

Table 2 presents the most frequent cybersecurity topic groups for the whole dataset. The results show that Transport Layer Security issues are on the top list of occurrences among all other cybersecurity problems following Cross-site scripting and SQL injection attacks. Transport Layer is the third layer of TCP/IP protocol (Transmission Control Protocol/ Internet Protocol). Cross-site scripting allows attackers to run client-side scripts from a website in the victim's client. In SQL injection attacks, malicious inputs can change the behaviour of an application that uses SQL in the back end to help the attacker gain unauthorised access or meddle with other security functions like authentication, confidentiality, anonymity, and privacy.

Table 3 presents the cybersecurity topic groups with the highest risk score among all the cybersecurity topics extracted with *CyRiPred*. Again, the first and third ranks are for Transport Layer Security issues and SQL injection attacks. But the second rank is for Adobe Flash Player

Table 2

Top ten most frequent cybersecurity topic groups among the CVEs in the NVD for all the years.

Rank	1	2	3	4	5	6	7	8	9	10
Most Frequent CAV Topics	Transport Layer Security	Cross-site scripting	SQL injection	List of TCP and UDP port numbers	Adobe Flash Player	IOS version history	Firefox version history	Cross-site request forgery	HTTP cookie	JavaScript

Table 3

Top ten riskiest cybersecurity topic groups among the CVEs in the NVD for all the years.

Rank	CAV topic	Impact (Average BaseScore)	Total	RiskScore
1	Transport Layer Security	6.19	3781	23 392.8
2	Adobe Flash Player	7.98	2862	22 832.3
3	SQL injection	6.62	3182	21051
4	List of TCP and UDP port numbers	5.89	3051	17 978.5
5	Cross-site scripting	4.81	3428	16 482.7
6	IOS version history	6.66	2356	15 686.4
7	Firefox version history	6.36	2107	13 397.6
8	Cross-site request forgery	5.4	1735	9371.9
9	Code injection	6.75	1070	7227.5
10	JavaScript	5.55	1282	7119.5

Table 4

Top ten most frequent cybersecurity topic groups for 2019, 2020 and prediction for 2021. In the evaluation section of the study, we predict 2019 and 2020 against actual observations.

Rank	2019	2020	2021 (level change from previous year)
1	List of TCP and UDP port numbers	List of TCP and UDP port numbers	List of TCP and UDP port numbers
2	IOS version history	ZFS (Not in the prediction for the next year)	IOS version history (+3)
3	Adobe Flash Player	Windows Vista	Firefox (+1)
4	Firefox	Firefox	Adobe Flash Player (New)
5	Adobe ColdFusion	IOS version history	Transport Layer Security (+4)
6	Transport Layer Security	Android (operating system)	SQL injection (+1)
7	List of HTTP status codes	SQL injection	Windows Vista (−4)
8	SQL injection	Cracking of wireless networks (Not in the prediction for the next year)	Adobe ColdFusion (+2)
9	Cross-site scripting	Transport Layer Security	Android (operating system) (−3)
10	Intel Active Management Technology	Adobe ColdFusion	Cross-site scripting (New)

vulnerabilities and exposures. We do not present a table for the top ten most severe cybersecurity topic groups because there are more than ten topic groups with the highest score of severity (*baseScore*) which is 10.

In Table 4, the most frequent cybersecurity topics are presented for the last two years and also a prediction for the next year, 2021. Expected changes in the ranking for the year 2021 compared to its previous year are also shown in parenthesis. If a cybersecurity topic goes out of the top ten list in the predicted year, it is mentioned in the parenthesis. The Zettabyte File System (ZFS) and “Cracking of wireless networks” are not in the predicted top ten frequent cybersecurity topics for 2021. The ZFS is a file system developed by Sun Microsystems, with the latest release in 2018. The top rank in 2019 and 2020 and the predicted top risk for 2021 is “List of TCP and User Datagram Protocol (UDP) port numbers”. This is precisely the title of the most similar Wikipedia article to the CVEs in these years. TCP and UDP ports are abstract connections in layer 3 of TCP/IP. Each port can be assigned to a particular application that needs communication with another client or server. UDP is a sub-protocol in layer 3 of TCP/IP for more error-tolerant connections with less delay. It should also be mentioned that topics like “IOS version history” or “Firefox version history” are automatically extracted for CVEs that talk about iOS or Firefox security related issues in their description field.

Table 5 shows the top ten cyber risks for the last two years (2019–20), and predictions for 2021 are presented. Given our time series of risk scores for each cybersecurity risk for each month from 1988 to 2020, the random forest regressor predicts the cyber risk scores of these cybersecurity topics for 2021. The random forest regressor is chosen due to its highest accuracy on average for all the cybersecurity topic

time series. The last column of Table 5 is the main output of *CyRiPred*. Windows Vista is on the list of top ten cybersecurity risks for 2021.

Legacy systems such as Windows Vista are those systems that are old and not updated and supported by the developer company anymore, but they are still in use. Adobe Flash Player and Adobe Cold Fusion (a software for rapid web development) are in the list of predicted top ten cyber risks of 2021 by *CyRiPred*. *CyRiPred* also predicts that ZFS and wireless cracking will drop off the top ten risks in 2021.

Moreover, the first rank for the last two years and the predicted year will be for “List of TCP and UDP ports”. This means that attacks on layer three ports will be the highest risk in the future year. The difference between “Transport Layer Security” and “List of TCP and UDP ports” is that the first one is more general. Attacks, vulnerabilities, and exposures are related to layer three (transport layer) but not necessarily ports.

5. Evaluation of the model

In the evaluation phase for *CyRiPred*, first, we tested two sub-modules of the *CyRiPred* model: *WikiTopic* and Random Forest regressor. Then, we evaluated the model by comparing its outputs to the results presented in different cybersecurity trend reports mentioned in the Discussion section. For *WikiTopic*, the topic extractor module, we chose 100 random articles from four other Wikipedia alternatives and their title as our test target labels.

The alternatives to Wikipedia (alternative online open-source encyclopedias to gather labelled test data) used in the evaluation were

Table 5

Top ten cyber risks for 2019, 2020, and prediction for 2021.

Rank	2019	2020	2021 (level change from previous year)
1	List of TCP and UDP port numbers	List of TCP and UDP port numbers	List of TCP and UDP port numbers
2	IOS version history	Windows Vista	IOS version history (+2)
3	Firefox version history	ZFS (Not in the prediction for the next year)	Firefox version history (+2)
4	Adobe ColdFusion	IOS version history	Adobe Flash Player (New)
5	Adobe Flash Player	Firefox version history	Adobe ColdFusion (New)
6	Transport Layer Security	SQL injection	Windows Vista (−4)
7	Intel Active Management Technology	List of Qualcomm Snapdragon systems-on-chip	Transport Layer Security (+3)
8	SQL injection	Android (operating system)	SQL injection (−2)
9	List of HTTP status codes	Cracking of wireless networks	Android (operating system) (−1)
		Not in the prediction for the next year)	
10	Android (operating system)	Transport Layer Security	List of Qualcomm Snapdragon systems (−3)

Table 6

Parameters of the base models for prediction.

Model	Parameters
Random Forest	n_estimators = 150
MLP Regressor	(31-7-3-1) - Activation = Logistic_Sigmoid, Learning_Rate = 0.001, Momentum = 0.9
KNN Regressor	K = 7
ARIMA	Best fit of p, d, q for each time series in the validation set
Support Vector Regressor	Kernel = RBF, Regularisation = 1.0

Scholarpedia, Citizendium, Britannica, and Infoplease³. We chose two random paragraphs from each article. Then, we used *WikiTopic* to extract the topic of these articles (two paragraphs as a delegate for the whole article) and finally calculated the accuracy of the *WikiTopic*. We sampled the articles because, in our application, the description data can be considered a short text (up to one paragraph). We used two sample paragraphs from each article in our test because some parts of the articles, like the history part, are unrelated to the article's topic. From 100 test articles, 96 articles were labelled correctly. The topic from Wikipedia was the same as the one in the alternative encyclopedia. This indicated an accuracy of 97% for the *WikiTopic* algorithm.

For the prediction module of *CyRiPred*, time series are partitioned into train, validation, and test sets for prediction evaluation. Monthly aggregated data (summation for topic occurrence and average for topic severity) starts from October 1988; the final data is for September 2020. Therefore, the time series will have 354 elements for each series for 354 months. Our prediction model tries to predict 15 months after that to have 12 months of the year 2021. We used 2017 and 2018 as a validation set for parameter selection and 2019 and 2020 for testing. After testing different popular supervised regressors like MLP (Multi-Layer Perceptron), KNN (K-Nearest Neighbour), SVR (Support Vector Regression), RF (Random Forest), and ARIMA (Auto-Regressive Integrated Moving Average), the best results in terms of MSE (Mean Square Error), MAE (Mean Absolute Error), and R squared were for Random Forest (in average for all the topic time series).

Our model's approach toward cyber risk prediction is unique; Therefore, we cannot compare our model to any existing model in the literature. However, we compare Random Forest as the prediction core of our model to other base supervised learners. The models' configurations are presented in Table 6. These configurations have been set using exhaustive grid searches in validation sets.

The MSE (Mean Squared Error), MAE (Mean Absolute Error), and RMSE (Root Mean Squared Error) results are presented in Table 7 for comparison. Formulas for these evaluation criteria are shown in Eq. (7). The MSE and RMSE raise the magnitude of significant errors and lessen

Table 7

Prediction results of different base models.

Model	MSE	MAE	RMSE	R2
Random Forest Regressor	7.868	0.457	2.805	0.981
MLP Regressor	11.923	0.781	3.452	0.970
KNN Regressor	17.123	0.923	4.137	0.965
ARIMA	24.167	0.998	4.916	0.962
Support Vector Regressor	10.188	0.556	3.192	0.979

the importance of minor errors due to squaring in their formulas. At the same time, MAE treats significant or insignificant errors the same. The R^2 criterion shows how well the data fit in a regression model.

$$\begin{aligned}
 MAE &= \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \\
 MSE &= \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad RMSE = \sqrt{MSE} \\
 R^2 &= 1 - \frac{(y_i - \hat{y}_i)^2}{(y_i - \bar{y})^2} \\
 \hat{y} &: PredictedValue \\
 \bar{y} &: MeanValue
 \end{aligned} \tag{7}$$

Evaluation criteria for the RF model with 150 regressor trees (150 estimators) were MAE = 0.457, MSE = 7.869, R Squared = 0.982). Finally, after choosing RF as our best fit model, we trained the model again with all 354 data to predict the next 15 elements (3 months in 2020 and 12 months in 2021). Table 8 presents the evaluation results of two main components of *CyRiPred*.

Table 9 is presented to show how significantly different our Random Forest model results are compared to other models. We used the Diebold–Mariano test for statistical significance results. The Diebold–Mariano test is the best choice for comparing the RMSE, MSE, or MAE of different prediction models. In our case, we compared the RMSEs of other models to the Random Forest model, which showed the best results in the test set. In our test, the H_0 hypothesis meant no difference between RMSEs of Random Forest and any other prediction models. At the same time, H_1 implied a significant difference between the RMSEs. In our Diebold–Mariano test, we assumed that a P -Value less than 0.05 meant that the difference between RMSEs was statistically significant.

³ Scholarpedia: <http://www.scholarpedia.org> Citizendium: <https://www.infoplease.com>, Britannica: <https://www.britannica.com>, Infoplease: <https://www.infoplease.com>

Table 8

Test results for WikiTopic topic extraction algorithm and random forest as the best predictor among different regressors.

WikiTopic Topic Extractor			
WikiTopic	Test Size 100 Test Articles with known topic	Alternative Encyclopedias Accuracy = 96%	Wikipedia Accuracy = 97%
Predictor (Test results for 2019–2020)			
Random Forest Regressor (150 Estimators)	MSE = 7.869	MAE = 0.457	R-Squared = 0.982

Table 9

P-values of the Diebold–Mariano test between the random forest model and other models.

Model	p-value
MLP Regressor	< 0.01
KNN Regressor	< 0.01
ARIMA	< 0.01
Support Vector Regressor	0.042 < 0.05

One can read the work of Diebold (2015) for a detailed explanation of the Diebold–Mariano test.

For a more qualitative testing approach for the whole *CyRiPred* model, we excluded the data's last two years, 2019, 2020 and tried to predict cyber risk scores for the cybersecurity topic groups. We saw that the ranks did not change for the top ten cyber risks, and there was also no error in rank prediction for high-occurrence cybersecurity topics.

6. Discussion

We first consider how much the predictions of *CyRiPred* are similar to the predictions of cyber risk companies.

Many annual cyber risk reports include a prediction for the most critical cyber risks of the following year. Our model does this job entirely data-driven without referring to experts. This benefits the cybersecurity and insurance companies like all those cyber risk reports. The difference between the output results of our model and those reports is that our model's output is expert-bias-free, and the procedure is clear, comprehensive and complete. Cybersecurity and cyber insurance companies can rely on the results of our model to make their future policies better than before when they know what they should expect in the future more accurately.

A comprehensive study by GovTech⁴ investigates emerging technologies and cybersecurity risks over 20 different cybersecurity trend prediction reports for 2020. GovTech included the reports of famous companies like Kaspersky, McAfee, AT&T, Gartner, Forbes, and Bit-Defender. Their reports and other reports like those for Cybriant⁵ and Cybercrime Magazine⁶ focus on 2021, and their predictions are a mixture of *CyRiPred*'s predictions and their risk perception.

We calculated each topic's average polar sentiment score (instead of *baseScore*, which measured severity in CVSS) to confirm. The sentiment score can indicate risk perception (Chang & Wang, 2018; Gaspar et al., 2016; Wu et al., 2014). Sentiment scores were derived with the Vader Sentiment algorithm (Gilbert & Hutto, 2014). The worst risk perceptions for 2020 and 2021 were Ransomware and Denial-of-service attacks (Rank first and second). The other cybersecurity topics within the top ten worst risk perceptions that were not in the top ten high-risk cybersecurity topics were VirtualBox, File inclusion vulnerability, Java, WordPress, and Heartbleed, an OpenSSL cryptography bug.

Ransomware, which has the worst risk perception, were among half of cybersecurity trend predictions for 2020. Other cybersecurity high-risk topics predicted with *CyRiPred* were among the trend prediction reports of cybersecurity companies. Our predictions are automatically generated and are data-driven, showing the risk perception versus reality issue. We emphasise that our model can predict cyber risk perception just by changing the CVSS baseScore to polar sentiment score.

Statista and IBM report that the most common cybersecurity attacks companies experience are phishing. Attacks like phishing need social engineering techniques where a user is persuaded to open a link. The NVD dataset records vulnerabilities and exposures of IT systems and software. Therefore, *CyRiPred* cannot make a forecast in the areas where the vulnerability is in the user.

We investigate the completeness of the *CyRiPred* model from two different aspects: First, we use a comprehensive risk assessment framework named CURF (Core Unified Risk Framework) (Wangen, 2017; Wangen et al., 2018). Then, we investigate the completeness of the vulnerabilities and exposures dataset and the knowledge repository we use as the input of our model. Regarding the CURF cyber risk completeness framework, our model fulfils all the primary activities and parts of the secondary activities of CURF. Here, the primary activities of CURF are mentioned with fulfilled secondary activities in parenthesis inside double quotes: Risk Identification (Assigning risks to vulnerabilities and exposures under "Vulnerability" secondary activity), Risk Estimation ("Probability estimation" using the data-driven occurrence calculation and "impact estimation" using CVSS dataset impact scores), and part of Risk Evaluation ("Risk prioritisation" by ranking risks). Our model mainly lacks the "Risk Treatment" secondary activity from the Risk Evaluation primary activity of the completeness evaluation framework. The "Risk Treatment" was outside the scope of our model, which was built mainly to be a machine learning-based data-driven prediction model.

Regarding the completeness of the datasets we used, it is worth mentioning that the CVE dataset is the most comprehensive repository of vulnerabilities and exposures, continuously updated by the National Cyber Security Division of the U.S. Department of Homeland Security. Wikipedia is chosen as the knowledge repository of cyber risk labelling. Wikipedia is also the largest continuously updating open-source encyclopedia in the world (Zhao & Strotmann, 2021). All these measures guarantee that the completeness of the model will be achieved to the highest standards possible in cyber risk recognition and prediction.

In our CyPiPred framework, the WikiTopic extraction algorithm operates as a sequential process designed meticulously to ensure thoroughness and accuracy in topic extraction. We acknowledge that in today's age of real-time processing, sequential operations may raise concerns about efficiency. However, it is imperative to understand the contextual application of this algorithm. The CyPiPred model is conceptualised as a real-time cyber risk prediction tool and a periodic reporting mechanism. Its primary purpose is to produce comprehensive cyber risk rankings that organisations can utilise to assess and adjust their cybersecurity postures. Given this objective, the model is intended to be run on a scheduled basis, such as monthly, aligning with typical enterprise risk reporting cycles. In such a scenario, the time taken by the WikiTopic algorithm becomes less critical as the emphasis shifts towards accuracy and depth of topic extraction. In essence, while the speed of data processing is undeniably valuable, the

⁴ GovTech cybersecurity trends report: <https://www.govtech.com/blogs/lohrmann-on-cybersecurity/the-top-20-security-predictions-for-2020.html> Online report, [last accessed: 1th Jan. 2021]

⁵ Cybriant: <https://cybriant.com/plan-today-for-cybersecurity-trends-in-2021/> Online report, [last accessed: 1th Jan. 2021]

⁶ Cybercrime Magazine: <https://cybersecurityventures.com/top-5-cybersecurity-facts-figures-predictions-and-statistics-for-2019-to-2021/> Online report, [last accessed : 1th Jan. 2021]

objective of the CyPiPred model emphasises the quality and reliability of the extracted topics over sheer processing speed. This approach, though seemingly counter-intuitive in real-time, aligns more closely with organisational cybersecurity management's strategic, longer-term decision-making processes.

We also highlight the problem of legacy systems that shows itself in the results of *CyRiPred*. The reader can mention cybersecurity topic groups for systems and applications that have been obsolete for years (legacy systems). This does not mean they do not create new security problems, but the fact that they are no longer supported can make them riskier. Windows Vista topic in recent years is an example of this in the results tables produced by *CyRiPred*.

The synergy between data-driven models and expert opinions is not overlooked in the quest for comprehensive cyber risk prediction. The recent work by Ji et al. (2020) sheds light on a bi-objective optimisation model to aggregate expert opinions, enhancing objectivity and reliability in group decision-making. Their approach revolves around representing experts' views as probability density functions and focusing on the consensus and confidence levels for more grounded predictions. Such a fusion of algorithmic predictions with expert insights, encapsulating objective and subjective perspectives, can be a powerful tool to apprehend unexpected risks. While our model emphasises the algorithmic paradigm, future iterations might benefit from integrating these expert opinion frameworks, offering a more holistic approach to cyber risk prediction.

7. Conclusion

This study provides a general model for identifying major cyber risk groups from the most comprehensive database of security vulnerabilities. In addition to identifying and classifying cyber risks, our model predicts future risks based on the extracted time series. Using out-of-sample techniques, we demonstrate the efficacy of our model and further confirm this with detailed robustness checks.

This research overcomes the inefficiency of previous models to provide a comprehensive method for recognising and predicting cyber risks. Our *CyRiPred* model does not have pre-assumptions and is not limited to specific applications, environments, or architectures.

As a suggestion for future research, extracting more complex structures compared to a simple classification of cyber risks from CVEs can be considered. Multi-level taxonomies can be extracted, and their evolution studied. This can potentially uncover the process of the creation of new cyber risks. By availability of more data in future, using different deep-learning models can be another choice to evaluate higher-performance models. In our research, we tried to show "HOW" our method predicts the cyber risks. As another suggestion for future research, interested researchers can think of "WHY"s instead of "HOW"s, which means discussing why a cybersecurity topic rank fluctuates in time. In light of our findings and the ever-evolving landscape of cyber threats, a logical next step would development of robust policies tailored to mitigate the highlighted risks. Given our model's focus on predicting cyber risks through CVE data, policy development can leverage these predictions to formulate proactive strategies that preemptively address potential vulnerabilities. This would bridge the gap between predictive analytics and actionable measures, offering a more comprehensive approach to cyber risk management. Exploring this synergy between prediction and policy could usher in a new era of cyber resilience, further enhancing the security of digital infrastructures.

Abbreviation

Application Programming Interface (API), Auto-Regressive Integrated Moving Average (ARIMA), Core Unified Risk Framework (CURF), Common Vulnerabilities and Exposures (CVE), Common Vulnerabilities Scoring System (CVSS), Common Weakness Enumeration (CWE), Cyber Risk Recognition and Prediction Model (*CyRiPred*), Decision

Support System (DSS), JSON (JavaScript Object Notation), K-Nearest Neighbour (KNN), Latent Dirichlet Allocation (LDA), Latent Semantic Analysis (LSA), Machine Learning (ML), Mean Absolute Error (MAE), Multilayer Perceptron (MLP), Mean Squared Error (MSE), Naïve Bayes (NB), National Institute of Standards and Technology (NIST), Natural Language Processing (NLP), National Vulnerabilities Database (NVD), Probabilistic latent semantic analysis (PLSA), Support Vector Machine (SVM), Support Vector Regression (SVR), Transmission Control Protocol/Internet Protocol (TCP/IP), Term Frequency/Inverse Document Frequency (TF/IDF), User Datagram Protocol (UDP), eXtensible Markup Language (XML), Yet Another Keyword Extraction (YAKE), Zettabyte File System (ZFS)

CRedit authorship contribution statement

Arash Negahdari Kia: Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft, Visualization. **Finbar Murphy:** Conceptualization, Methodology, Investigation, Writing – review & editing. **Barry Sheehan:** Conceptualization, Methodology, Investigation, Writing – review & editing. **Darren Shannon:** Conceptualization, Methodology, Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

CVE data is publicly available.

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